- Computational Fluid Dynamics - (CFD)

It is the art of replacing the integrals or the partial differential terms in the equations, derived from fundamental principles governing fluid flow, with discretized algebraic forms which in turn are solved to obtain numbers for the flow field values at discrete points in time and/or space.

3 fundamental principles governing fluid flow are -

(i) Mass is conserved
(ii) Newton's second law
(iii) Energy is conserved

} Governing Eq^n's ⇒ { Continuity Eq^n, Momentum Eq^n, Energy Eq^n.

Application of CFD -

(i) It is applied for design and simulation purposes.

(ii) It is applied in case of Automobile & Engine -

- Study external flow over body of vehicle
- " internal flow through engine.

It helps automotive engineers to better understand the physical flow processes and in turn to design improved vehicle.

(iii) Industrial Manufacturing Applications

CFD acts as a research tool and contribute information of direct application to manufacturing.

(iv) Civil Engg. Application -

Problems involving rheology of rivers, lakes, estuaries, etc. are all the subject of investigations using CFD.

(v) Environmental Engg. Applications -

The discipline of heating, air conditioning and circulation through building have all come under application of CFD.

eg. It provides information for design of furnaces with increased thermal efficiency and reduced emmission of pollutants.

(vi) Naval Architecture Applications -

CFD is a major tool in solving hydro-dynamics problem associated with ships, submarines torepedos etc.

—x——x—

$\lim_{t_2 \to t_1} \frac{\rho_2 - \rho_1}{t_2 - t_1} = \frac{D\rho}{Dt}$ = Instantaneous time rate of change of density.

$\frac{D}{Dt}$ = Substantial derivative

$\frac{\partial \rho}{\partial t}$ = time rate of change of density at fixed point.

Substantial derivative for cartesian coordinate:

$$\frac{D}{Dt} = \frac{\partial}{\partial t} + u\frac{\partial}{\partial x} + v\frac{\partial}{\partial y} + w\frac{\partial}{\partial z}$$

$\therefore$ Vector operator, $\nabla = i\frac{\partial}{\partial x} + j\frac{\partial}{\partial y} + k\frac{\partial}{\partial z}$

Hence $\boxed{\frac{D}{Dt} = \frac{\partial}{\partial t} + (V \cdot \nabla)}$ → Valid for any coordinate system.

- $\frac{D}{Dt}$ → Substantial Derivative
- $\frac{\partial}{\partial t}$ → Local Derivative
- $(V \cdot \nabla)$ → Convective derivative

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Physical Meaning

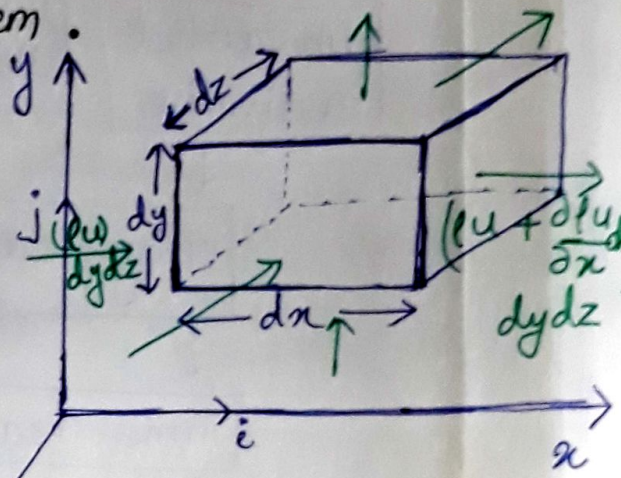
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Time rate of change

- Substantial Derivative → following a moving fluid element
- Local Derivative → at a fixed point
- Convective derivative → due to movement of fluid element from one location to another in the flow field where flow properties are spatially different.

Continuity Eqⁿ – (Mass is conserved)

Consider infinitesimally element of side dx, dy and dz in x, y, z direction respectively in a cartesian coordinate system.



As the mass flow through this element, the velocity and density which are function of spatial location changes.

Hence mass flow across opposite faces along a direction changes.

Hence,

$\rho = \rho(x, y, z, t)$

$v = ui + vj + wk$

$u = u(x, y, z, t)$

$v = v(x, y, z, t)$

$w = w(x, y, z, t)$

Then, net outflow in x, y, z direction are respectively –

$$\left[\rho u + \frac{\partial \rho u}{\partial x} dx\right] dydz - (\rho u) dydz = \frac{\partial (\rho u)}{\partial x} dxdydz \quad \text{(i)}$$

$$\left[\rho v + \frac{\partial \rho v}{\partial y} dy\right] dxdz - (\rho v) dxdz = \frac{\partial (\rho v)}{\partial y} dxdydz \quad \text{(ii)}$$

$$\left[\rho w + \frac{\partial \rho w}{\partial z} dz\right] dxdy - (\rho w) dxdy = \frac{\partial (\rho w)}{\partial z} dxdydz$$

∴ Net Mass flow out of element =

$$\left[\frac{\partial (\rho u)}{\partial x} + \frac{\partial (\rho v)}{\partial y} + \frac{\partial (\rho w)}{\partial z}\right] dxdydz$$

Total mass of fluid in infinitesimally element = $\rho(dx\,dy\,dz)$

Time rate of mass increase = $\frac{\partial \rho}{\partial t}(dx\,dy\,dz)$

But, Net mass flow out of element = time rate of decrease of mass inside

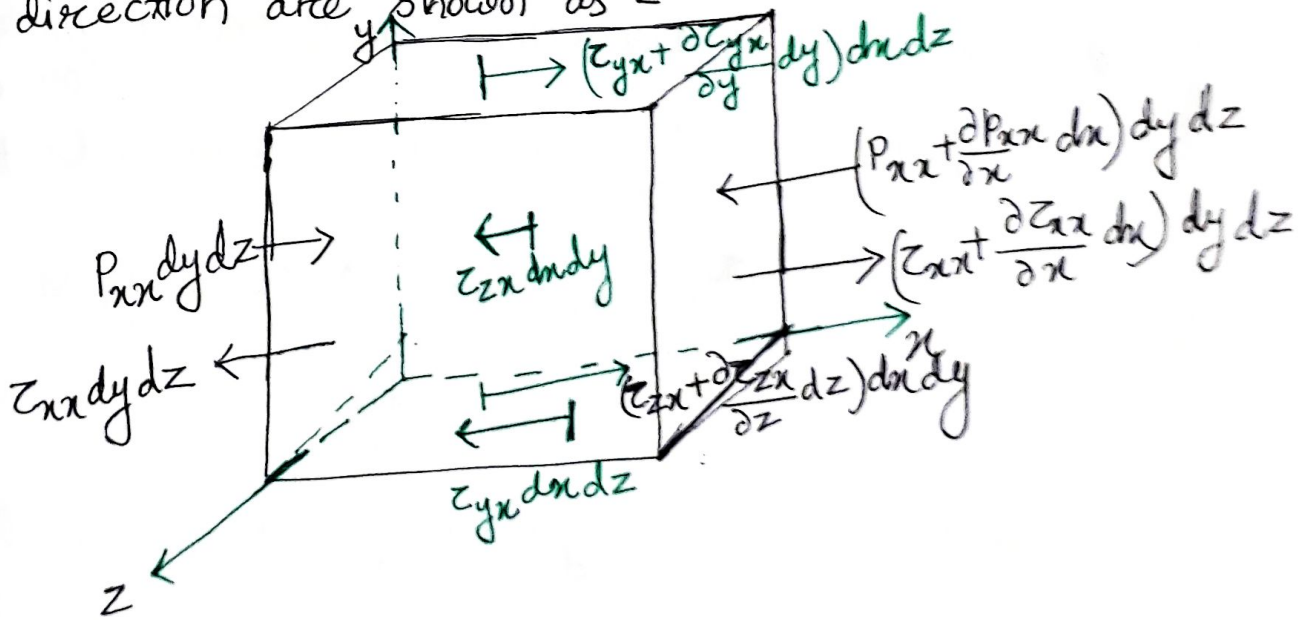
$$\Rightarrow \left[\frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} + \frac{\partial(\rho w)}{\partial z}\right] dx\,dy\,dz = -\frac{\partial \rho}{\partial t}(dx\,dy\,dz)$$

$$\Rightarrow \boxed{\frac{\partial \rho}{\partial t} + \left[\frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} + \frac{\partial(\rho w)}{\partial z}\right] = 0}$$

$$\Rightarrow \boxed{\frac{\partial \rho}{\partial t} + \nabla(\rho \mathbf{V}) = 0} \quad \text{Conservation form}$$

The Momentum Eqn —
(Newton's 2nd law — F = ma)

Consider a infinitesimally small, moving fluid element and the forces acting only in the x-direction are shown as —



Considering all forces in x-direction only -

$F_x = m a_x$ [Newton's 2nd law]

Let Body force / unit mass acting on fluid element be f, with f_x as its x component.

Volume of fluid element = $dx\,dy\,dz$

Hence, Body forces on fluid element acting in x-direction = $\rho f_x (dx\,dy\,dz)$

Shear and normal stress depends on time rate of change of the deformation of fluid element along x-direction.

Net surface force in x-direction.

$$= \left[p - \left(p + \frac{\partial p}{\partial x} dx\right)\right] dy\,dz + \left[\left(\tau_{xx} + \frac{\partial \tau_{xx}}{\partial x} dx\right) - \tau_{xx}\right] dy\,dz +$$

$$\left[\left(\tau_{yx} + \frac{\partial \tau_{yx}}{\partial y} dy\right) - \tau_{yx}\right] dx\,dz + \left[\left(\tau_{zx} + \frac{\partial \tau_{zx}}{\partial z} dz\right) - \tau_{zx}\right] dx\,dy$$

Total force in x direction = Body force + Surface force

$$(F_x) = -\frac{\partial p}{\partial x} dx\,dy\,dz + \frac{\partial \tau_{xx}}{\partial x} dx\,dy\,dz +$$

$$\frac{\partial \tau_{yx}}{\partial y} dx\,dy\,dz + \frac{\partial \tau_{zx}}{\partial z} dx\,dy\,dz +$$

$$\rho f_x\, dx\,dy\,dz$$

$$\Rightarrow m a_x = \left(-\frac{\partial p}{\partial x} + \frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z}\right) + \rho f_x\, dx\,dy\,dz$$

$\therefore$ $\Rightarrow \rho\, dx\,dy\,dz\, a_x = \rho\, dx\,dy\,dz$

$a_x = \frac{Du}{Dt}$ $\Rightarrow \rho \frac{Du}{Dt} = -\frac{\partial p}{\partial x} + \frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z} + \rho f_x$

So, x-component of momentum eqn for a viscous flow,

$$\rho \frac{Du}{Dt} = -\frac{\partial p}{\partial x} + \frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z} + \rho f_x$$

|| y-component

$$\rho \frac{Dv}{Dt} = -\frac{\partial p}{\partial y} + \frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \tau_{yy}}{\partial y} + \frac{\partial \tau_{zy}}{\partial z} + \rho f_y$$

z-component –

$$\rho \frac{Dw}{Dt} = -\frac{\partial p}{\partial z} + \frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \tau_{yz}}{\partial y} + \frac{\partial \tau_{zz}}{\partial z} + \rho f_z$$

→ Navier Stokes eqn in non-conservation form.

In terms of substantial Derivative,

$$\rho \frac{Du}{Dt} = \rho \frac{\partial u}{\partial t} + \rho V \cdot \nabla u \quad \text{——(i)}$$

Expanding,

$$\frac{\partial(\rho u)}{\partial t} = \rho \frac{\partial u}{\partial t} + u \frac{\partial \rho}{\partial t}$$

Now Rearranging (i)

$$\rho \frac{\partial u}{\partial t} = \frac{\partial(\rho u)}{\partial t} - u \frac{\partial \rho}{\partial t} \quad \text{—(ii)}$$

Using vector identity for the divergence of the product of a scalar times a vector, we get

$$\nabla \cdot (\rho u V) = u \nabla(\rho V) + \rho V \nabla u$$

$$\Rightarrow \rho V \nabla u = \nabla \cdot (\rho u V) - u \nabla(\rho V) \quad \text{—(iii)}$$

Now Eqn (i) becomes -

$$\rho \frac{Du}{Dt} = \frac{\partial(\rho u)}{\partial t} - u\frac{\partial \rho}{\partial t} - u\nabla(\rho V) + \nabla(\rho u V)$$

$$= \frac{\partial(\rho u)}{\partial t} - \left[u\frac{\partial \rho}{\partial t} + \vec{u}^{\,0}\nabla(\rho V)\right] + \nabla(\rho u V)$$

$$\boxed{\rho \frac{Du}{Dt} = \frac{\partial(\rho u)}{\partial t} + \nabla(\rho u V)}$$

Derivation of Energy Equation :-

According to 1st law of thermodynamics :-

Rate of change of energy inside the fluid element = Net flux of heat into the element + Rate of workdone on element due to body and surface forces

$\Rightarrow$ A = B + C

where A = Rate of change of energy inside the fluid element.

B = Net flux of heat into the element

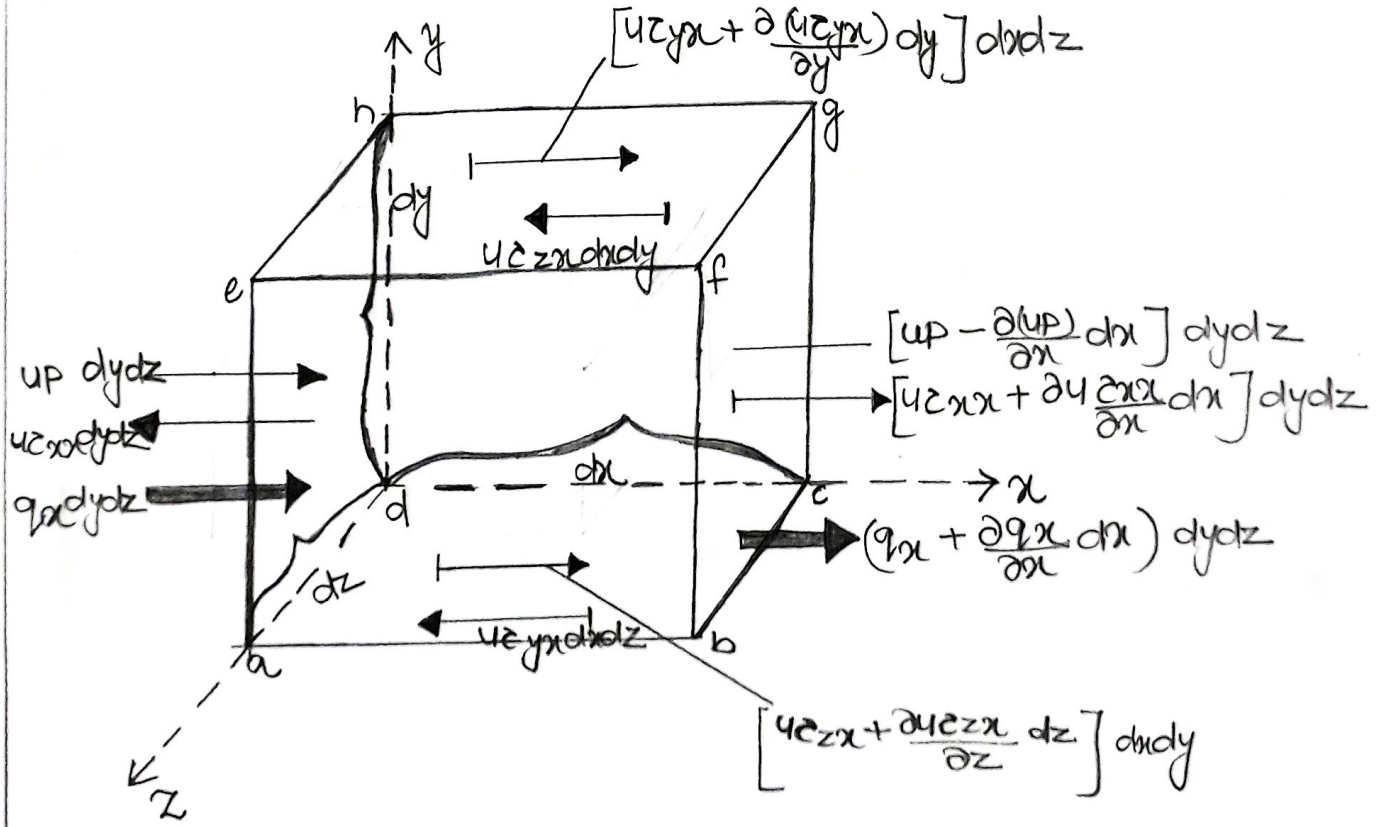
C = Rate of workdone on element due to body and surface forces.

Let us first evaluate 'C'.

So here workdone by the body force and surface forces are to be evaluated.

Hence the rate of workdone by the body force acting on the fluid element moving at a velocity v is $\rho f \cdot v(dxdydz) - (1)$

Let us consider the surface forces (pressure force and shear force, normal forces) in x-direction



Net pressure force on x- direction $= \left[up - \left(up + \frac{\partial up}{\partial x} dx\right)\right] dydz$

$$= -\frac{\partial(up)}{\partial x} dx\, dy\, dz$$

Net shear forces on x- direction

$$= \left[-\tau_{yx} + \tau_{yx} + \frac{\partial(\tau_{yx})}{\partial z} dz\right] dxdy + \left[-\tau_{zx} + \tau_{zx} + \frac{\partial(\tau_{zx})}{\partial y} dy\right] dxdz$$

$$= \frac{\partial(\tau_{yx})}{\partial z} dxdydz + \frac{\partial(\tau_{zx})}{\partial y} dxdydz$$

Net Normal force on x-direction

$$= -\tau_{xx}\, dydz + \left[\tau_{xx} + \frac{\partial \tau_{xx}}{\partial x} dx\right] dydz$$

$$= \frac{\partial \tau_{xx}}{\partial x} dx\, dy\, dz$$

So the net rate of work done on the moving fluid element due to these forces is simply

$$\left[-\frac{\partial(up)}{\partial x} + \frac{\partial(\tau_{xx})}{\partial x} + \frac{\partial(\tau_{yx})}{\partial y} + \frac{\partial(\tau_{zx})}{\partial z}\right] dx\, dy\, dz \quad -(ii)$$

Similarly we will get the result in y and z direction.

So, considering those we can say

$$c = \left\{-\left[\frac{\partial(up)}{\partial x} + \frac{\partial(vp)}{\partial y} + \frac{\partial(wp)}{\partial z}\right] + \left[\frac{\partial(\tau_{xx})}{\partial x} + \frac{\partial(v\tau_{yy})}{\partial y} + \frac{\partial(w\tau_{zz})}{\partial z}\right.\right.$$
$$+ \frac{\partial(\tau_{yx})}{\partial y} + \frac{\partial(\tau_{zx})}{\partial z} + \frac{\partial(v\tau_{xy})}{\partial x} + \frac{\partial(v\tau_{zy})}{\partial z} + \frac{\partial(w\tau_{xz})}{\partial x}$$
$$\left.\left. + \frac{\partial(w\tau_{yz})}{\partial y}\right]\right\} dxdydz + \int f \cdot v\, dx\, dy\, dz$$

Now lets evaluate the term B.

B = Net flux of heat into the element

This flux is due to (1) volumetric heating such as absorption or emission of radiation and (2) heat transfer across the surface due to temp. gradients i-e thermal condition.

Let q' = rate of volumetric heat addition per unit mass.

So, volumetric heating of element = $\rho q' dx\, dy\, dz$ —(III)

Heat transfered by the thermal condition into the moving fluid element across the left face of the fig. = $q_x\, dy\, dz$, where q_x is the heat transfferred through the right face of the fig

$= \left[q_x + \frac{\partial q_x}{\partial x} dx\right] dy\, dz$

So, Net heat transfered in x- direction into the fluid element by thermal condition is

$$q_x - \left[q_x + \frac{\partial q_x}{\partial x} dx\right] dy\, dz = -\frac{\partial q_x}{\partial x} dx\, dy\, dz$$

Taking into account heat transfer in the y & z direction across other faces in the drawn fig.

Heating of fluid element by thermal condition $= -\left(\frac{\partial q_x}{\partial x} + \frac{\partial q_y}{\partial y} + \frac{\partial q_z}{\partial z}\right) dx\, dy\, dz$ —(IV)

The term B is the sum of eq^n (III) + eq^n (IV)

$$B = \left[\rho q - \left(\frac{\partial q_x}{\partial x} + \frac{\partial q_y}{\partial y} + \frac{\partial q_z}{\partial z}\right)\right] dx\, dy\, dz$$

$$q_x = -k\frac{\partial T}{\partial x}, \quad q_y = -k\frac{\partial T}{\partial y}, \quad q_z = -k\frac{\partial T}{\partial z}$$

$$\text{So, } B = \left[\rho q + \frac{\partial}{\partial x}\left(k\frac{\partial T}{\partial x}\right) + \frac{\partial}{\partial y}\left(k\frac{\partial T}{\partial y}\right) + \frac{\partial}{\partial z}\left(k\frac{\partial T}{\partial z}\right)\right] dx\, dy\, dz$$

Let us evaluate the term A

we know that the fluid element has two contributions to its energy.

1. The ~~thermal~~ internal energy ~~into~~ due to random molecular motion, e (per unit mass)
2. The kinetic energy due to translational motion of the fluid element. The kinetic energy per unit mass is simply $v^2/2$.

Hence the moving fluid element has both internal and kinetic energy. The sum of these two is the total energy.

or, mathematically A, $e + \frac{v^2}{2}$ = total energy

Now the time rate of change of total energy per unit mass is given by the substantial derivative. Since the mass of the fluid element is $\rho\, dx\,dy\,dz$, we have

$$A = \rho \frac{D}{Dt}\left(e+\frac{V^2}{2}\right) dx\,dy\,dz \quad \text{—(III)}$$

eq^n(III) = eq^n(I) + eq^n(II)

$$\Rightarrow \rho \frac{D}{Dt}\left(e+\frac{V^2}{2}\right) dx\,dy\,dz = \left[\rho q + \frac{\partial}{\partial x}\left(k\frac{\partial T}{\partial x}\right) + \frac{\partial}{\partial y}\left(k\frac{\partial T}{\partial y}\right) + \frac{\partial}{\partial z}\left(k\frac{\partial T}{\partial z}\right)\right.$$
$$- \frac{\partial}{\partial x}(up) - \frac{\partial(vp)}{\partial y} - \frac{\partial}{\partial z}(wp) + \frac{\partial}{\partial x}(u\tau_{xx}) +$$
$$\frac{\partial}{\partial y}(u\tau_{yx}) + \frac{\partial}{\partial z}(u\tau_{zx}) + \frac{\partial}{\partial x}(v\tau_{xy}) + \frac{\partial}{\partial y}(v\tau_{yy})$$
$$+ \frac{\partial}{\partial z}(v\tau_{zy}) + \frac{\partial}{\partial x}(w\tau_{xz}) + \frac{\partial}{\partial y}(w\tau_{yz}) +$$
$$\left.\frac{\partial}{\partial z}(w\tau_{zz}) + \rho\cdot f\cdot v\right] dx\,dy\,dz \quad \text{—(IV)}$$

we know that

$$u\left[\rho\left(\frac{Du}{Dt}\right)\right] = \left[-\frac{\partial p}{\partial x} + \frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + \frac{\partial}{\partial z}\tau_{zx} + \rho f_x\right] u$$

$$\Rightarrow \rho \frac{D(u^2/2)}{Dt} = -u\frac{\partial p}{\partial x} + u\frac{\partial \tau_{xx}}{\partial x} + u\frac{\partial \tau_{yx}}{\partial y} + u\frac{\partial}{\partial z}\tau_{zx} + \rho u f_x \quad \text{—(V)}$$

Like wise for y and z-direction

$$\frac{\rho D(v^2/2)}{Dt} = -v\frac{\partial p}{\partial y} + v\frac{\partial \tau_{xy}}{\partial x} + v\frac{\partial \tau_{yy}}{\partial y} + v\frac{\partial \tau_{zy}}{\partial z} + \rho v f_y \quad \text{—(VI)}$$

$$\frac{\rho D(w^2/2)}{Dt} = -w\frac{\partial p}{\partial z} + w\frac{\partial}{\partial y}\tau_{xz} + w\frac{\partial \tau_{yz}}{\partial y} + w\frac{\partial}{\partial z}\tau_{zz} + \rho w f_z \quad \text{—(VII)}$$

again $u^2 + v^2 + w^2 = v^2$

so adding eq^n(V) + eq^n(VI) + eq^n(VII) we get

$$\rho \frac{D(v^2/2)}{Dt} = -u\frac{\partial p}{\partial x} - v\frac{\partial p}{\partial y} - w\frac{\partial p}{\partial z} + u\left[\frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z}\right] + v$$
$$\left(\frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \tau_{yy}}{\partial y} + \frac{\partial \tau_{zy}}{\partial z}\right) + w\left(\frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \tau_{zy}}{\partial y} + \frac{\partial}{\partial z}\tau_{zz}\right)$$
$$+ \rho(uf_x + uf_y + wf_z) \quad \text{—(VIII)}$$

Now subtracting eqn (VIII) from eqn (IV) we have

$$\rho \frac{De}{Dt} = pq + \frac{\partial}{\partial x}\left(k\frac{\partial T}{\partial x}\right) + \frac{\partial}{\partial y}\left(k\frac{\partial T}{\partial y}\right) + \frac{\partial}{\partial z}\left(k\frac{\partial T}{\partial z}\right) - p\left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z}\right) + \tau_{xx}\frac{\partial u}{\partial x} + \tau_{yx}\frac{\partial u}{\partial y} + \tau_{zx}\frac{\partial u}{\partial z} + \tau_{xy}\frac{\partial v}{\partial x} + \tau_{yy}\frac{\partial v}{\partial y} + \tau_{zy}\frac{\partial v}{\partial z} + \tau_{xz}\frac{\partial w}{\partial x} + \tau_{yz}\frac{\partial w}{\partial y} + \tau_{zz}\frac{\partial w}{\partial z} \quad -(IX)$$

[Note :- $\rho f.v = \rho(uf_x + vf_y + wf_z)$]

Equation (IX) can also be written as

$$\rho \frac{De}{Dt} = pq + \frac{\partial}{\partial x}\left(k\frac{\partial T}{\partial x}\right) + \frac{\partial}{\partial y}\left(k\frac{\partial T}{\partial y}\right) + \frac{\partial}{\partial z}\left(k\frac{\partial T}{\partial z}\right) - p\left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z}\right) + \tau_{xx}\frac{\partial u}{\partial x} + \tau_{yy}\frac{\partial v}{\partial y} + \tau_{zz}\frac{\partial w}{\partial z} + \tau_{yx}\left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}\right) + \tau_{zx}\left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x}\right) + \tau_{zy}\left(\frac{\partial v}{\partial z} + \frac{\partial w}{\partial y}\right) \quad -(X)$$

Eqn (X) is known as energy equation.